

Yousef Saleh

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Mechanical Engineer | Research and Development Test & Validation | Thermal/Fluid Systems | Instrumentation & Automation

PROFESSIONAL PROFILE

Mechanical engineer with a doctorate in Mechanical Engineering from Florida State University and experience across research and development test systems, compressor/chiller validation, thermal-fluid analysis, instrumentation, calibration/metrology, automation, experimental aeroacoustics, and controlled technical documentation. Turns complex lab work into repeatable execution by planning calibrations, troubleshooting measurements, automating analysis, strengthening documentation, coordinating vendors, and coaching engineers so tests run consistently. Research background adds high-pressure/high-temperature supersonic testing, custom hardware, acoustic mapping, synchronized data acquisition, and publication-quality analysis.

EDUCATION

Doctor of Philosophy, Mechanical Engineering | Florida State University | May 2026

Dissertation: [Aeroacoustics of non-axisymmetric supersonic jets](#)

Master of Science, Mechanical Engineering | Florida State University | Fall 2023

Bachelor of Science, Aerospace Engineering | The University of Texas at Austin | December 2018

TECHNICAL CAPABILITIES

- **Thermal and fluid systems:** heat transfer, thermodynamics, refrigeration fundamentals, compressor performance, pressure-ratio behavior, corrected mass-flow analysis, high-pressure and high-temperature systems, supersonic flow
- **Test and validation:** hardware validation, test-platform readiness, calibration planning, measurement systems analysis, measurement traceability, instrumentation, data acquisition, root-cause analysis, controlled documentation, audit support
- **Automation and analysis:** Python, MATLAB, LabVIEW, compressor maps, compressor performance curves, efficiency contours, digital signal processing, image-based automation prototypes, executable packaging, tools for technicians and engineers
- **Hardware and controls:** industrial controllers, compact real-time controllers, protocol gateways, networked communications hardware, electrical standards, pressure standards, temperature references, power meters, flow meters
- **Quality and process:** root-cause analysis, failure-mode analysis support, controlled procedures, calibration standards, training records, document control, measurement traceability workflows, vendor coordination
- **Design and experimental methods:** SOLIDWORKS and computer-aided design, nozzle design, custom mechanisms, fixture concepts, schlieren, shadowgraph, particle image velocimetry, acoustic measurement, experimental facility operation

PROFESSIONAL EXPERIENCE

Senior Lab Engineer | Danfoss Turbocor

Jan 2025 - Present

- Lead validation readiness across compressor and chiller research and development test platforms, coordinating calibration planning, troubleshooting, measurement systems analysis, controlled documentation, and lab support.
- Build Python-based compressor analysis tools for performance curves, speedline maps, efficiency contours, corrected mass-flow trends, and anomaly review.
- Develop and refine calibration workflows for pressure, temperature, electrical, flow, and torque instrumentation.
- Troubleshoot thermocouple configuration, pressure signals, power meters, data acquisition behavior, serial and network communications, and measurement traceability.
- Build semi-automated electrical calibration tools for technicians and engineers; harden them with clearer error reporting, channel checks, input validation, and improved user guidance.
- Advance pressure and temperature calibration automation from prototype scripts toward deployable tools, including executable packaging, training, and handoff support.
- Build and verify a duplicate portable hardware test platform integrating an industrial controller, compact real-time controller, data acquisition, protocol gateway, power distribution, and network hardware.
- Complete electrical calibration training and apply it to internal electrical calibration procedures, handheld instrument troubleshooting, and lab environmental awareness.

- Establish and expand internal torque-wrench calibration capability through documentation, training, standard setup review, measurement-limit discussions, and spare-parts coordination.
- Coordinate external vendors and internal stakeholders for pressure standards, power-meter calibration, water-flow-meter selection, chamber service, lab build-out, and lab readiness.

Graduate Research Engineer | Florida State University

Aug 2019 - Jan 2025

- Designed and executed high-pressure, high-temperature aeroacoustic experiments, operating supersonic facilities up to 1500 K stagnation temperature and Mach 2.5 fully expanded jet conditions.
- Operated a high-pressure anechoic facility for supersonic jets, non-axisymmetric nozzle geometries, acoustics, flow visualization, and thermal-fluid behavior.
- Designed custom nozzles, precision mechanisms, and experimental hardware for controlled supersonic testing and large-scale data collection.
- Designed an in-situ rotating nozzle mechanism indexed in 0.25-degree increments and automated Method of Characteristics workflows for supersonic nozzle development and acoustic mapping.
- Calibrated and synchronized eight precision microphones and data systems to improve repeatability and directional acoustic measurements.
- Automated acquisition and processing workflows for large experimental datasets, including acoustic reconstruction, instrumentation synchronization, data reduction, and Python/MATLAB analysis.
- Built flow-visualization methods including shadowgraph, schlieren, particle image velocimetry, and a Z-type schlieren system.
- Defined fuel- and thermal-system components for a hot-jet retrofit and mentored students through setup, testing, analysis, and technical communication.

Research Assistant | The University of Texas at Austin

Dec 2016 - Dec 2018

- Improved data reliability in a Mach 1.8 wind-tunnel environment by reconfiguring acquisition workflows and preventing data loss during testing.
- Installed hardware and developed a feedback control solution to regulate shock location in a scramjet isolator under incoming perturbations.
- Built hardware and software applications to interpret flow properties and analyze physical behavior in high-speed test environments.

SELECTED TECHNICAL PROJECTS

- **Compressor Performance Analysis:** Developed Python-based compressor performance workflows for compressor performance curves, efficiency contours, pressure-ratio behavior, corrected mass-flow analysis, and review of anomalous compressor datasets.
- **Calibration Automation:** Built pressure, temperature, and electrical calibration automation workflows using electrical standards, power meters, image-based automation prototypes, networked communications, and deployment considerations for tools used by technicians and engineers.
- **Portable Test Platform Communications and Data Cluster:** Built a duplicate hardware platform integrating an industrial controller, compact real-time controller, data acquisition, a protocol gateway, power distribution, and networking hardware for prototype bring-up and system replication.
- **Water Flow Meter Replacement:** Derived requirements, coordinated vendor discussions, and supported fit checks, calibration support sessions, hardware inspection, and troubleshooting for new flow meters.
- **Heated Jet Thermal Retrofit:** Defined fuel and thermal system components in piping and instrumentation format, including heating and cooling requirements for a heated supersonic jet system.
- **Shock-Location Control Study:** Supported closed-loop control research by installing hardware and developing feedback-control logic to regulate shock-train location under perturbations.

PUBLICATIONS

- Saleh, Y. (2026). *Aeroacoustics of non-axisymmetric supersonic jets*. Doctoral dissertation, Florida State University. <https://doi.org/10.17605/OSF.IO/X5CE6>
- Reyes, A. N., Saleh, Y., Gustavsson, J., Jolowsky, C. N., Kumar, R., Treadwell, L. J., & Sweat, R. D. (2024). *Supersonic hot jet ablative testing and analysis of boron nitride nanotube hybrid composites*. *Composites Part B: Engineering*, 284, 111684.
- Reyes, A., Treadwell, L., Saleh, Y., Gustafson, J., Kumar, R., & Sweat, R. (2023). *Supersonic Hot Jet Analysis of Boron Nitride Nanotube Hybrid Composites Under Extreme Heat and Pressure Conditions*. CAMX 2023.
- Saleh, Y., Sellappan, P., & Alvi, F. S. (2023). *Experimental Characterization of the Acoustics of a Supersonic Diamond Jet*. *AIAA Journal*, 61(10), 4546–4555.
- Saleh, Y., Seckin, S., Song, M., & Alvi, F. S. (2023). *Mapping 3D Acoustics of a Supersonic Diamond Jet: An Odyssey in the Golden Age of Aeroacoustics*. AIAA AVIATION 2023 Forum.
- Prasad, A., Saleh, Y., Sellappan, P., Unnikrishnan, S., & Alvi, F. S. (2022). *Effects of Expansion Ratio and Nozzle Asymmetry on Flowfield of Diamond Jets*. *AIAA Journal*, 60(9), 5215–5231.
- Saleh, Y., & coauthors. (2021). *Experimental and Computational Study of a Mach 2 Diamond Jet*. AIAA SCITECH 2021 Forum.
- Vanstone, L., Bosco, A., Saleh, Y., Akella, M., Clemens, N. T., & Gogineni, S. (2020). *Closed-loop control of unstart in a Mach 1.8 isolator*. *Journal of Propulsion and Power*, 36(1), 153–157.

TEACHING, TRAINING, AND MENTORSHIP

- Developed and delivered internal instruction on refrigeration and thermodynamics fundamentals for engineers and test operators.
- Built custom slide decks, examples, quizzes, and assessments tied to compressor and chiller testing so the material was usable in practice.
- Mentored interns, technicians, junior engineers, and students in calibration execution, experimental setup, data processing, documentation, troubleshooting, and safe lab execution.
- Built structured training plans and task lists so new team members could contribute safely and consistently.
- Supported a Florida A&M University–Florida State University capstone collaboration through project proposal, kickoff, lab tours, scheduling, simplified piping and instrumentation work, and scope development.

LEADERSHIP AND SERVICE

Chief Engineer, Aircraft Design | The University of Texas at Austin

Jan 2018 - May 2018

- Led a team of 8 engineers through aircraft design requirements, risk assessment, sizing, payload, power, and control decisions.
- Evaluated control surfaces, electrical connections, wiring, sensors, and autopilot systems.
- Performed risk assessment and generated mitigation protocols for an unmanned air system concept.
- Developed an emergency-response concept for surveillance and payload drops.

President, Water Ski Team | The University of Texas at Austin

Nov 2016 - Nov 2017

- Coordinated 60 members and a 10-person officer board.
- Oversaw a \$55,000 operating budget covering competition lake lease, equipment, boat maintenance, travel, lodging, and merchandise.
- Established a modern website, updated team branding, and developed long-term growth and talent-development plans.
- Recruited coaching support and refocused team operations toward competitiveness and member development.

CERTIFICATIONS AND TRAINING

- EPA Section 608 Universal
- Fluke MET 101 electrical calibration training
- Fluke MET 301 advanced electrical calibration training
- Fluke MET 302 uncertainty analysis in electrical calibration training
- Arc flash safety training
- Lockout/tagout training
- Compressor instruction and in-person compressor training
- Measurement systems analysis (MSA)
- Export control, internal compliance, annual EHS, and responsible business conduct training